

Financial Time Series

46-929

Homework #5

Due: Friday, April 30, 2021, 1:30pm.

1. GARCH-in-Mean

The text by Hamilton (p667) states “Finance theory suggests that an asset with a higher perceived risk would pay a higher return on average. For example, let Y_t denote the ex-post rate of return on some asset minus the return on a safe alternative asset. Suppose that Y_t is decomposed into a component anticipated by investors at time $t - 1$ (denoted by m_t) and a component that was unanticipated (denoted by a_t):

$$Y_t = m_t + a_t.$$

Then the theory suggests that the mean return (m_t) would be related to the variance of the return σ_t^2 . In general, the ARCH-M (GARCH-M) called ARCH-in-mean (GARCH-in-mean) is given by:

$$Y_t = \beta_0 + \beta_1 X_t + \delta \sigma_t^2 + a_t,$$

$$a_t = \sigma_t \epsilon_t,$$

$$\sigma_t^2 = 1 + 0.4a_{t-1}^2.$$

Here the $\{\epsilon_t$ are i.i.d. $N(0,1)$ random variables.” The ARCH/GARCH-in-mean model has the additional feature that the conditional volatility appears in the mean model. Assume that $\beta_0 = .06$, $\beta_1 = 0.30$, and $\delta = 0.20$, and the process $\{X_t\}$ is the inflation rate during this time period.

- a) What is $E(Y_t|X_t = 0.1, a_{t-1} = 0.6)$?
 - b) What is $Var(Y_t|X_t = 0.1, a_{t-1} = 0.6)$?
 - c) Is the conditional distribution of Y_t given X_t and a_{t-1} normal? Why or why not?
 - d) Is the marginal distribution of Y_t normal? Why or why not?
2. Download stock data Merck (ticker = MRK) from 1/1/2011 to 4/23/2021. Run through a complete analysis of time series for the log-returns of this data series.
- (a) Plot the log-returns and analyze the mean model. Check for stationarity using the adjusted Dickey Fuller test. Make any necessary transformations to make the series stationary.
 - (b) Look at the ACF and PACF plots and test for any significant autocorrelation using the Ljung-Box test. Report your results.
 - (c) If there are any significant autocorrelations, find a good ARMA model for these data.
 - (d) Test for any ARCH effects using the Ljung-Box test and the Engle test.
 - (e) If there are any significant ARCH effects, fit a GARCH model to the data.
 - (f) Using the data up to 4/1/21, forecast the volatility going forward to 4/23/2021.
 - (g) Suppose you own 100 shares of MRK on 4/1/2021. What is a 99% confidence interval for the price of MRK on 4/23/2020 based on the models you have developed?